

L7a: SIM Portfolios and a Risk-Free Asset

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Objectives for Today

By the end of this lecture, you should be able to:

- **Compare minimum-variance portfolios with SIM and data-driven inputs:** identify the co-movement the SIM leaves out, and compare the allocations under a common covariance and on held-out prices.
- **Construct the capital allocation line and tangent portfolio:** derive the line relating expected growth to risk, identify its slope as the Sharpe ratio, and find the risky portfolio that maximizes it.
- **Choose and evaluate a complete portfolio:** explain lending, borrowing, and financing limits, and compare complete portfolios with an index fund.

Lectures and Examples

Lecture: SIM Portfolios and a Risk-Free Asset

Example 1: Compare Data-Driven and SIM Portfolios

Compare covariances, frontiers, and weights, then test on 2025 prices.

Example 2: Construct the Tangent Portfolio and Capital Allocation Line

Does every target pick the same risky portfolio? Test lending and borrowing in 2025.

Both examples run on the firms chosen in the L6b client interview, or on thirteen default firms. The L6b examples show how we estimated the SIM parameters.

Concept Review: The SIM Covariance

In the SIM $g_i = \alpha_i + \beta_i g_M + \varepsilon_i$, residuals have zero mean and are uncorrelated with the market and across assets, so $\mathbb{E}[g_i] = \alpha_i + \beta_i \mathbb{E}[g_M]$. **On the diagonal:**

$$\text{Var}(g_i) = \beta_i^2 \text{Var}(g_M) + 2\beta_i \underbrace{\text{Cov}(g_M, \varepsilon_i)}_{=0} + \text{Var}(\varepsilon_i) \quad \text{expand the SIM}$$

$$= \beta_i^2 \sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2 \quad \text{second assumption}$$

Off the diagonal ($i \neq j$), the second and third assumptions remove three terms:

$$\begin{aligned} \text{Cov}(g_i, g_j) &= \beta_i \beta_j \text{Var}(g_M) + \beta_i \underbrace{\text{Cov}(g_M, \varepsilon_j)}_{=0} + \beta_j \underbrace{\text{Cov}(\varepsilon_i, g_M)}_{=0} + \underbrace{\text{Cov}(\varepsilon_i, \varepsilon_j)}_{=0} \\ &= \beta_i \beta_j \sigma_{g,M}^2 \end{aligned}$$

Concept Review: SIM Portfolio Inputs

With the betas in β , split each entry into its market and residual parts:

$$\Sigma_g = \sigma_{g,M}^2 \begin{bmatrix} \beta_1^2 & \beta_1\beta_2 & \cdots \\ \beta_2\beta_1 & \beta_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} + \begin{bmatrix} \sigma_{g,\varepsilon,1}^2 & 0 & \cdots \\ 0 & \sigma_{g,\varepsilon,2}^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} = \underbrace{\sigma_{g,M}^2 \beta\beta^\top}_{\text{market}} + \underbrace{\mathbf{D}_g}_{\text{residual}}$$

Substituting the training-data estimates gives the SIM inputs:

$$\hat{\mathbf{g}}_{\text{SIM}} = \hat{\alpha} + \hat{\beta} g'_M, \quad \hat{\Sigma}_{g,\text{SIM}} = s_{g,M}^2 \hat{\beta}\hat{\beta}^\top + \hat{\mathbf{D}}_g$$

- **Same means:** a least-squares line with an intercept passes through the sample means, so $\hat{\alpha}_i + \hat{\beta}_i g'_M = g'_i$.
- **Different covariance:** off the diagonal, the SIM covariance is the sample covariance with the residual covariances set to zero.

Portfolio Risk under a SIM

For fixed weights $\mathbf{w}$, substituting the SIM entries of Σ_g into the portfolio variance gives:

$$\begin{aligned}\text{Var}(g_p) &= \sum_i \sum_j w_i w_j \text{Cov}(g_i, g_j) && \text{portfolio variance (L6a)} \\ &= \sum_i w_i^2 (\beta_i^2 \sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2) + \sum_i \sum_{j \neq i} w_i w_j \beta_i \beta_j \sigma_{g,M}^2 && \text{SIM entries} \\ &= \sigma_{g,M}^2 \sum_i \sum_j (w_i \beta_i)(w_j \beta_j) + \sum_i w_i^2 \sigma_{g,\varepsilon,i}^2 && \text{collect market terms} \\ &= \sigma_{g,M}^2 \left(\sum_i w_i \beta_i \right)^2 + \sum_i w_i^2 \sigma_{g,\varepsilon,i}^2 && \text{a perfect square} \\ &= \underbrace{\beta_p^2 \sigma_{g,M}^2}_{\text{market risk}} + \underbrace{\sum_i w_i^2 \sigma_{g,\varepsilon,i}^2}_{\text{residual risk}} && \beta_p = \sum_i w_i \beta_i\end{aligned}$$

Market risk depends on $\mathbf{w}$ only through β_p . Even weights over many assets with uncorrelated, similar-size residuals drive residual risk toward zero.

The Long-Only Problem with SIM Inputs

SIM-1 is L6a's long-only problem with SIM inputs and a floor $g_\star$ (year⁻¹):

$\begin{aligned} \underset{\mathbf{w}}{\text{minimize}} \quad & \underbrace{s_{g,M}^2 (\hat{\boldsymbol{\beta}}^\top \mathbf{w})^2}_{\text{market risk}} + \underbrace{\sum_i s_{g,\varepsilon,i}^2 w_i^2}_{\text{residual risk}} \\ \text{subject to} \quad & \hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w} \geq g_\star \\ & \mathbf{1}^\top \mathbf{w} = 1 \\ & 0 \leq w_i \leq 1, \quad i \in \mathcal{P} \end{aligned}$	$\begin{aligned} \text{minimize} \quad & \mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{g,\text{SIM}} \mathbf{w} \\ & \text{meet or exceed the growth target} \\ & \text{weights sum to one} \\ & \text{exclude short positions.} \end{aligned}$
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Sweeping $g_\star$ up from the long-only GMV growth rate traces the efficient frontier. We compare both allocations under one covariance, then on 2025 prices.

Example 1: [Compare Data-Driven and SIM Portfolios](#)

Company Profile: BlackRock

BlackRock is the world's largest investment manager, with \$15.3 trillion in client assets at the end of June 2026. Larry Fink and seven partners founded it in 1988 to manage fixed-income assets.

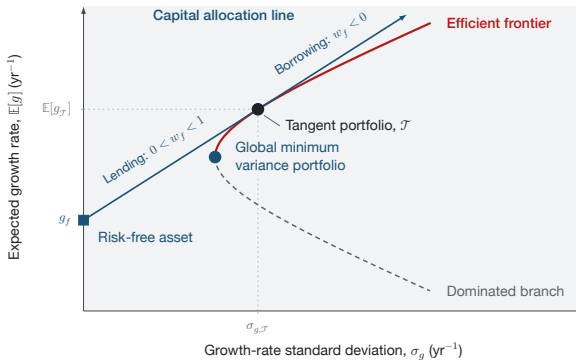
- **Index funds:** iShares came with Barclays Global Investors in 2009. Its indexing traces to John McQuown's 1971 Wells Fargo fund, one of the first.
- **Allocation funds:** iShares Core allocation ETFs run from 30% to 80% stocks.
- **Capital market assumptions:** the BlackRock Investment Institute publishes expected returns, volatilities, and correlations.
- **Aladdin:** built to manage portfolios and risk, and sold since 1999.

Today's connection: if every investor holds the same risky fund, together they hold the market, so a broad index fund approximates that fund.

Explore: [Student opportunities](#) [2027 Americas internship](#) [YouTube channel](#)

Adding a Risk-Free Asset

A **risk-free asset** has a known growth rate g_f (year^{-1}) and zero variance over the holding period, such as a Treasury bill held to maturity:



- **Capital allocation line (CAL):** mixing the risk-free asset with a risky portfolio gives a straight line from $(0, g_f)$ through that portfolio.
- **Tangent portfolio $\mathcal{T}$:** the steepest such line touches the efficient frontier at $\mathcal{T}$.
- **Lending and borrowing:** points before $\mathcal{T}$ lend at g_f , and points beyond it borrow at g_f .

The Capital Allocation Line

A **complete portfolio** c holds w_f in the risk-free asset and $1 - w_f$ in a risky portfolio p :

$$\begin{aligned}\mathbb{E}[g_c] &= w_f g_f + (1 - w_f) \mathbb{E}[g_p] && \text{expectation is linear} \\ &= g_f + (1 - w_f)(\mathbb{E}[g_p] - g_f) && \text{regroup}\end{aligned}$$

$$\sigma_{g,c}^2 = w_f^2 \underbrace{\text{Var}(g_f)}_{=0} + 2w_f(1 - w_f) \underbrace{\text{Cov}(g_f, g_p)}_{=0} + (1 - w_f)^2 \sigma_{g,p}^2 \quad g_f \text{ is constant}$$

$$\sigma_{g,c} = |1 - w_f| \sigma_{g,p} \quad \text{take the root}$$

For $w_f \leq 1$ and $\sigma_{g,p} > 0$, substituting $1 - w_f = \sigma_{g,c}/\sigma_{g,p}$ gives the **CAL**:

$$\mathbb{E}[g_c] = g_f + \text{SR}_p \sigma_{g,c}, \quad \text{SR}_p = (\mathbb{E}[g_p] - g_f)/\sigma_{g,p} \text{ (Sharpe ratio)}$$

We lend for $0 < w_f < 1$, hold p alone for $w_f = 0$, and borrow for $w_f < 0$.

Daily data: the slope is a one-day Sharpe ratio. With independent increments, $\text{SR}_p^{\text{annual}} = \sqrt{252} \text{SR}_p$ (Sharpe, 1994).

What Is in the Tangent Portfolio?

The **tangent portfolio** $\mathcal{T}$ is the fully invested risky portfolio with the largest Sharpe ratio. Let $\boldsymbol{\mu}_g$ hold the expected growth rates, $\boldsymbol{\Sigma}_g$ be positive definite, and $\mathbf{e} = \boldsymbol{\mu}_g - g_f \mathbf{1}$ be the excess-growth vector.

Short positions allowed: assume GMV growth (shorts allowed) exceeds g_f , and solve:

$$\underset{\mathbf{w}}{\text{maximize}} \quad \text{SR}(\mathbf{w}) = \frac{\mathbf{e}^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_g \mathbf{w}}} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \in \mathbb{R} \quad (\text{T-1})$$

Scaling $\mathbf{w}$ leaves the ratio unchanged, so we set its gradient to zero (T-2, [derivation](#)) and rescale. The weights point along $\boldsymbol{\Sigma}_g^{-1} \mathbf{e}$, and summing to one gives:

$$\boxed{\mathbf{w}_{\mathcal{T}}^{\text{short}} = \frac{\boldsymbol{\Sigma}_g^{-1} (\boldsymbol{\mu}_g - g_f \mathbf{1})}{\mathbf{1}^\top \boldsymbol{\Sigma}_g^{-1} (\boldsymbol{\mu}_g - g_f \mathbf{1})}} \quad (\text{T-3})$$

The GMV condition makes the denominator positive, so the CAL touches the efficient frontier.

Tangent Portfolio: Long-Only Risky Holdings

What changes if we cannot short the risky assets? We now require $w_i \geq 0$. The definition of the tangent portfolio is the same, but the feasible portfolios change:

$$\underset{\mathbf{w}}{\text{maximize}} \quad \frac{\mathbf{e}^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \Sigma_g \mathbf{w}}} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \geq 0 \quad (\text{T-LO})$$

We assume at least one risky asset has $\mu_{g,i} > g_f$, so a positive Sharpe ratio is attainable.

- **T-3 gives nonnegative weights:** it is feasible for T-LO and already maximizes the Sharpe ratio over the larger set of portfolios, so it is the long-only tangent portfolio.
- **T-3 gives some negative weights:** zeroing them and renormalizing is tempting, but it does not, in general, give the optimum. We must determine which assets receive zero weight and reoptimize the remaining holdings.

The worked example computes the long-only tangent weights using a convex minimum-variance problem.

Two-Fund Separation

The steepest CAL separates choosing a risky portfolio from choosing how much risk to take. Mean-variance investors with the same inputs who can lend and borrow freely at g_f hold the **same risky fund** $\mathcal{T}$, combined with the risk-free asset. Their risk preferences set w_f (Tobin, 1958).

Real investors face limits on how they finance their positions:

- **Long-only weights**: the risky fund is the long-only maximum-Sharpe portfolio, and investors still choose w_f .
- **No borrowing** ($w_f \geq 0$): the efficient boundary follows the CAL up to $\mathcal{T}$, then the risky efficient frontier.
- **Costlier borrowing**: a borrowing rate above g_f gives borrowers a flatter line through a different risky portfolio.

Risky and Risk-Free Allocation with SIM Inputs

Let's find the **long-only** tangent portfolio using the SIM estimates. Once we can lend or borrow, the risky weights no longer need to sum to one, so we keep track of two sets of weights:

- **Fund weights** $\mathbf{w}_{\mathcal{F}}$: the mix within the risky fund.
- **Wealth weights** $\mathbf{w}$: each risky holding as a fraction of initial wealth.

They differ by the **risky fraction** θ , the share of initial wealth held in risky assets:

$$\mathbf{w} = \theta \mathbf{w}_{\mathcal{F}}, \quad \mathbf{1}^{\top} \mathbf{w}_{\mathcal{F}} = 1, \quad \theta = \mathbf{1}^{\top} \mathbf{w} = 1 - w_f$$

For example, take a risky fund with weights (0.6, 0.4):

- **Lending** ($\theta = 0.5$): the wealth weights are (0.3, 0.2) and $w_f = 0.5$.
- **Borrowing** ($\theta = 1.2$): the wealth weights are (0.72, 0.48) and $w_f = -0.2$. We borrow 20% of initial wealth, while both risky holdings remain long positions.

Long-Only Holdings with Lending or Borrowing

Assume $\hat{\Sigma}_{g,\text{SIM}}$ is positive definite and at least one estimated mean growth rate exceeds g_f . For a feasible target $g_\star > g_f$ (year⁻¹), problem SIM-2 chooses $\mathbf{w}$ to solve:

minimize $\mathbf{w}^\top \hat{\Sigma}_{g,\text{SIM}} \mathbf{w}$	minimize growth-rate variance
subject to $(\hat{\mathbf{g}}_{\text{SIM}} - g_f \mathbf{1})^\top \mathbf{w} \geq g_\star - g_f$	meet the excess-growth target
$0 \leq w_i \leq 1, \quad i \in \mathcal{P}$	long-only holdings with per-asset caps.

- **No budget constraint:** we do *not* impose $\mathbf{1}^\top \mathbf{w} = 1$. The budget is $\mathbf{1}^\top \mathbf{w} + w_f = 1$, and w_f may be negative.
- **Bounds:** $w_i \geq 0$ prohibits shorting risky assets. The cap $w_i \leq 1$ is an additional limit that does not follow from the budget.
- **Solution:** SIM-2 is a convex quadratic program. We solve it numerically, and the solver determines which risky holdings are zero.

Continuation: Finding the Tangent Portfolio

A **continuation method** increases the target until the risk-free holding reaches zero. We lend when $\theta < 1$, borrow when $\theta > 1$, and hold the tangent portfolio at $\theta = 1$.

Initialize: set $g_\star^{(0)} = g_f$, $\mathbf{w}^{(0)} = \mathbf{0}$, $\theta^{(0)} = 0$, $g_L = g_f$, and $g_U = \max_i \hat{g}_{\text{SIM},i}$. Choose an increment $\Delta g > 0$ and a tolerance $0 < \epsilon < 1$. Then repeat:

1. **Check:** if $|\theta^{(k)} - 1| < \epsilon$, stop and return $\mathbf{w}_{\mathcal{T}} = \mathbf{w}^{(k)} / \theta^{(k)}$.
2. **Update the interval:** set $g_L = g_\star^{(k)}$ when lending ($\theta^{(k)} < 1$), otherwise $g_U = g_\star^{(k)}$.
3. **Next target:** step up to $\min(g_\star^{(k)} + \Delta g, g_U)$ until $\theta \geq 1$, then bisect $[g_L, g_U]$.
4. **Solve SIM-2** at the new target, starting from $\mathbf{w}^{(k)}$, and set $\theta^{(k+1)} = \mathbf{1}^\top \mathbf{w}^{(k+1)}$.

The highest-growth asset alone meets g_U with $\theta \geq 1$, so the sweep must cross $\theta = 1$. Past the tangent point, the risky mix stays the same until a per-asset cap binds.

Example 2: [Construct the Tangent Portfolio and Capital Allocation Line](#)

Optional Advanced Material

The [advanced index](#) links the companions:

- [Tangent portfolio derivation](#): the full calculation behind T-1 to T-3 and the connection to risky and risk-free allocation.
- [Estimation risk in portfolio weights](#): resample trading days, compare GMV and tangent weights, and test long-only allocations on 2025 prices.

Summary

- **The covariance model changes the allocation:** the SIM keeps the sample means but sets the residual covariances to zero. Under the sample covariance, the SIM weights carry at least as much risk at every growth floor.
- **The tangent portfolio sets the steepest CAL:** the Sharpe ratio is the line's slope, and maximizing it selects the tangent portfolio.
- **Lending and borrowing set the complete portfolio's risk:** investors who can lend and borrow freely at the risk-free rate hold the same risky fund. Financing limits can change that fund.

The tangent portfolio supplies the risky fund, and lending or borrowing sets the complete portfolio's risk.